MACHINE STRUCTURE 1

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Plan

- Introduction
- Chapter 1: Data coding and representation on machine
- Chapter 2: Boolean algebra
- Chapter 3: Combinatory logic

INTRODUCTION

- **Language**: a means of expressing and communicating
- **Our written language uses a code based on 26 letters (upper and lower case), 10 digits, punctuation symbols and mathematical signs.**

- Thanks to this code and these rules, we can transmit information, give instructions, count...

- computers are called "artificial intelligence", they have no ability to apprehend the outside world.

- Their only intelligence lies in their speed of execution of two-state order combinations (0 , 1)equivalent to (off , on).

- Coding consists of establishing a correspondence between the external representation of the information we use and its internal representation in the machine, which is a sequence of bits (a sequence of 0s and 1s).

Chapter 1:

Data coding and representation on machine

Chapter 1: Data coding and representation on machine

- a.1 Numeral systems
- a.2 Arithmetic Binary
- arithmetic operations
- b. Negative integers
- b.1 Representation of negative numbers in MSC (Module Sign Code)
- b.2 Representation of negative numbers in CP1
- b.3 Negative numbers in CP2
- a.2 Arithmetic
- C2 operations
- c. Real numbers
- c.1 Fixed-point representation of real numbers
- c.2 Floating-point representation of real numbers
- c.3 Arithmetic
- Floating-point operations
- Coding and representation –Digital
- a. ASCII code
- b. BCD code
- c. GRAY code
- d. Unicode

1. NUMERAL SYSTEM

- **Definition:**

  a set of symbols (numbers or letters) and writing rules for positioning these symbols.

Example: decimal number system, this system is composed of 10 digits: 0 1 2 3 4 5 6 7 8 9, each number is composed of a succession of digits and each digit has a weight.

Example: $4567 = 4 \times 10^3 + 5 \times 10^2 + 6 \times 10^1 + 7 \times 10^0$

the first digit on the left is the most significant digit

2-Bases

A B-base numeral system is defined by B symbols (numbers and/or letters). Let N be a number of n digits represented in base B.

- $N = a_{n-1}a_{n-2}\ldots a_i \ldots a_0 \quad \forall i \;\; a_i < B$ (All symbols are strictly less than B)
- Whatever the base, the polynomial form of N is :
- $N = a_{n-1} * B^{n-1} + a_{n-2} * B^{n-2} \ldots . + a_i * B^i \ldots . + a_0 * B^0$

3. BASE CHANGE

- From Base 10 to Base B
  Base B to base 10
- From base B1 to base B2
- From base 2 to base 8
- From base 8 to base 2
- From base 2 to base 16
- From base 16 to base 2
- From base 8 to base 16 or vice versa

3.1 From Base 10 to Base B

- Converting whole numbers

- The rule to follow is successive division

- Divide the number by B (base)

- Save the remainder, then divide the quotient by B, and so on until you obtain a quotient of zero.

- The sequence of remainders corresponds to the number in the target base

- The first remainder corresponds to the low weight(LSB) and the last to the high weight (MSB)

2. TYPES OF NUMBERING SYSTEMS

- **Binary system (Base 2) :**

uses 2 digits {0, 1}

- **Octal system (Base 8) :**
  uses 8 digits {0, 1, 2, 3, 4, 5, 6, 7}

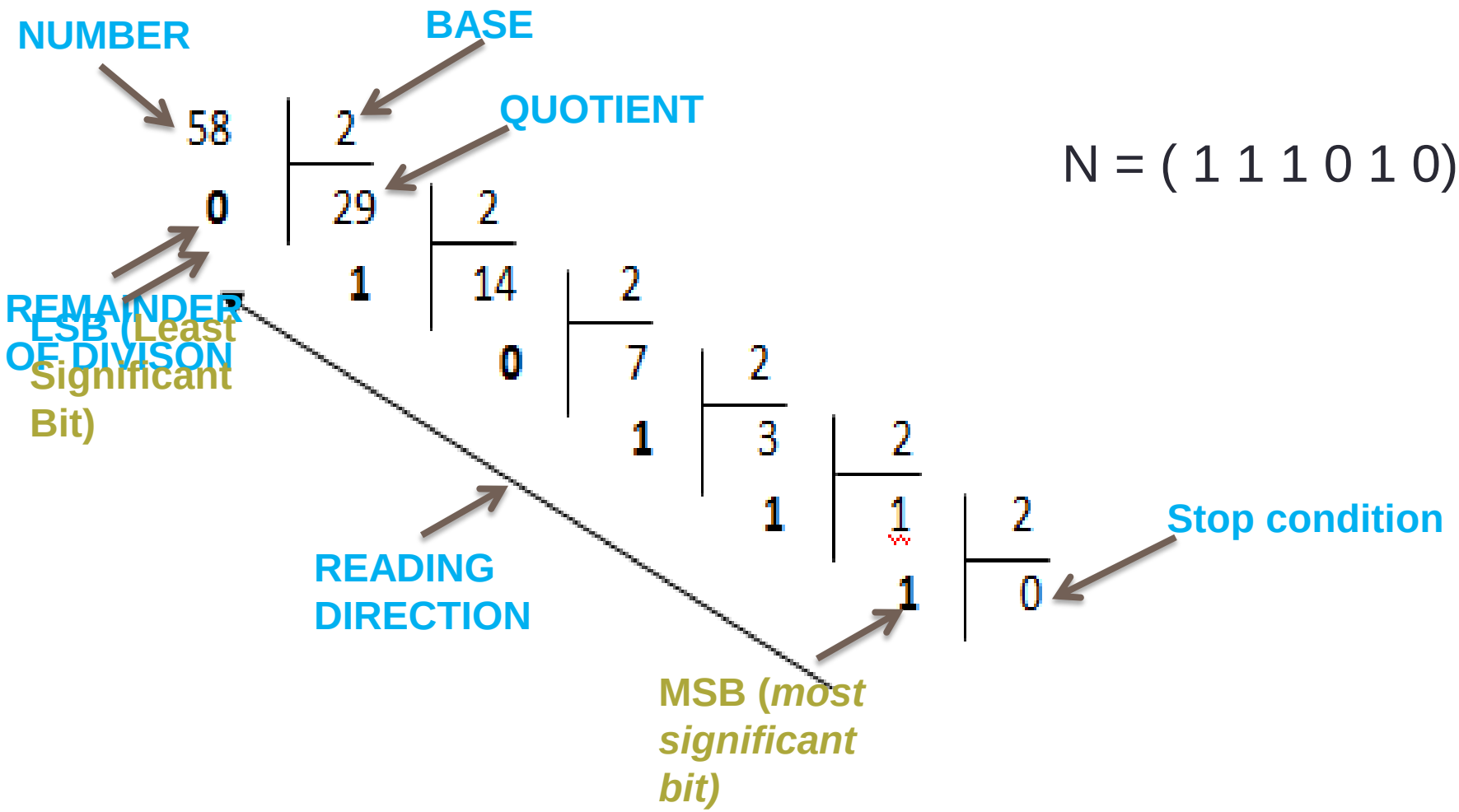
- **Hexadecimal system (Base 16) :**

uses 16 symbols (digits and letters): {0,1,2,3,4,5,6,7,8,9,A,B C,D, E, F}

A is the decimal equivalent of 10, B is the equivalent of 11...F is the equivalent of 15.

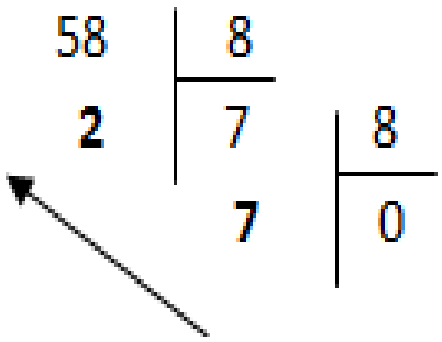
Examples :

- $(58)_{10} = (N)_2$



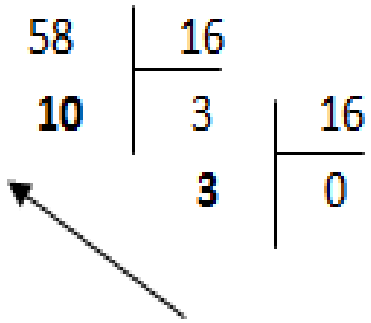
$N = (1\,1\,1\,0\,1\,0)_2$

- $(58)_{10} = (N)_8$



$N = (7\ 2)_8$

- $(58)_{10} = (N)_{16}$

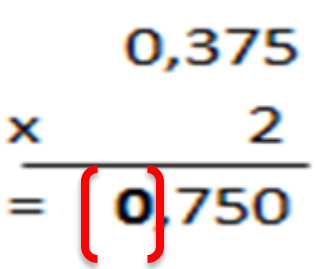


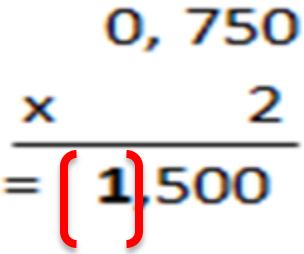
$N = (3\ A)_{16}$

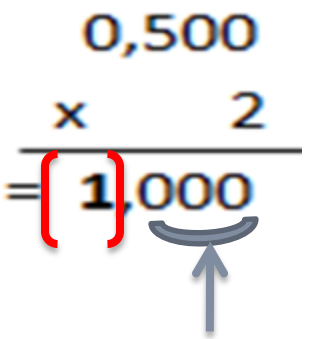
- Converting fractional numbers

- The integer part is treated as described above.
- For the decimal part, the rule is successive multiplication.
- Multiply the decimal part by B, then save the integer part of the result.
- Then multiply the new decimal part by B and so on until you obtain a zero decimal part.
- The sequence of integer parts obtained corresponds to the decimal part of the target base.

- **Example** 1: Convert $(58{,}375)_{10}$ to base 2







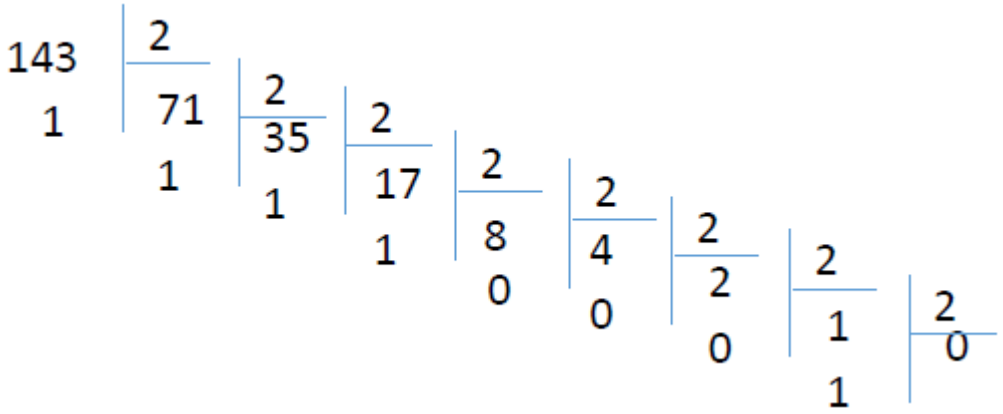
$(5\ 8)_{10} = (111010)_2$

$(0{,}375)_{10} = (0{,}011)_2$ so

$(5\ 8{,}375)_{10} = (111010{,}011)_2$

Example 2 (143,526)10 =(?)2

(143)10=(10001111)2



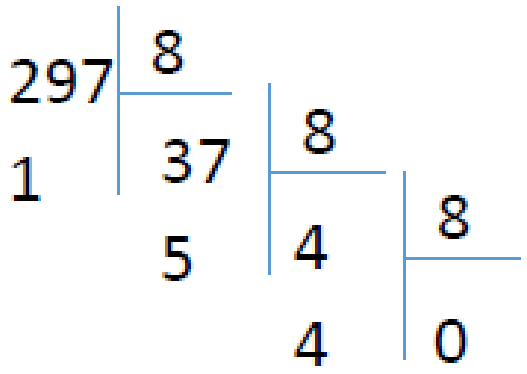
0,526 *2=**1**,052
0,052 *2=**0**,104
0,104 *2=**0**,208
0,208 *2=**0**,416
0,416 *2=**0**,832
0,832 *2=**1**,664
0,664 *2=**1**,328

(0,526)10=(0,1000011)2

(143,526)10=(10001111,1000011)2

Example 3: convert (297,358)10 to base 8

(297)10=(451)8



0,358 *8=**2**,864
0,864 *8=**6**,912
0,912 *8=**7**,296
0,296 *8=**2**,368
0,368 *8=**2**,944
0,944 *8=**7**,552
0,552 *8=**4**,416

Then (0,358)10=(0,2672274)8

So (297,358)10=(451,2672274)8

3.2 From Base B to Base 10

represent the number in polynomial form and calculate the sum of all terms.

$$N = a_{n-1} * B^{n-1} + a_{n-2} * B^{n-2} \ldots . + a_i * B^i \ldots . + a_0 * B^0$$

digit of number

Base

Number of bits(Weight)

Examples

- **$N = (1111011)_2 = (?)10$**

6 5 4 3 2 1 0

N= 1 1 1 1 0 1 1

$N = 1x2^6 + 1x2^5 + 1x2^4 + 1x2^3 + 0x2^2 + 1x2^1 + 1x2^0$

$N = 64 + 32 + + 16 + 8 + 0 + 2 + 1$

$N = (123)_{10}$

- **$N = (7B)_{16} = (?)10$**

1 0

N = 7 B

$N = 7x16^1 + 11x16^0$

$N = 112 + 11$

$N = (123)_{10}$

- **N=(1572)8 = (?)10**

3 2 1 0

N= 1 5 7 2

$N=2*8^0 +7*8^1+ 5*8^2+ 1*8^3$

$N=2 + 56+ 320+ 512$

$N=(890)10$

- Negative exponents are used for fractional numbers.

Example 1 : Convert $(7B,84)_{16}$ to base 10

1 0 -1 -2

N= 7 B, 8 4

$N = 7x16^{1} + 11x16^{0} + 8x16^{-1} + 4x16^{-2}$

$N = 112 + 11 + 0,5 + 0,015625 = (123,515625)_{10}$

Example 2

3 2 1 0 -1-2

Convert (1572,36)8 to base 10

$N=1*8^{3}+5*8^{2}+7*8^{1}+2*8^{0}+3*8^{-1}+6*8^{-2}=512+320+56+2+0,375+0,09375$

N=(890,46875)10

3.3 From Base B1 to Base B2

To switch from a B1 base to a B2 base, 2 operations are required.

- from base B1 to base 10
- then from base 10 to base B2

Example : Convert (3141)5 to base 16

(3141)5 = (421)10 (polynomial)

(421)10 = (1A5)16 (succ div)

(3141)5 = (1A5)16

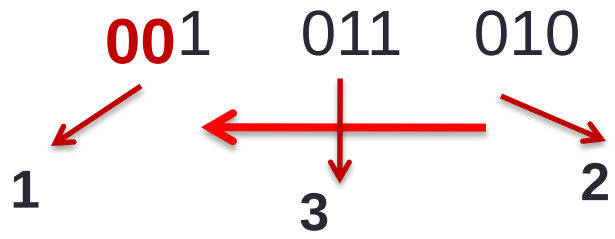
3.4 From base 2 to base 8

OCTAL	BINARY
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

- divide this number into groups of 3 bits and replace each group with its octal value (starting from the right).

- If the number of bits is not a multiple of 3, complete with 0's on the left.

- **Example** : Convert $(1011010)_2$ to base 8



$(001\ 011\ 010)_2 = (1\ 3\ 2)_8$

OCTAL	BINARY
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

3.5 From base 8 to base 2

- transcribe each digit of this number into 3 bits (starting from the left).
- **Example** : Convert $(645)_8$ to base 2

6 4 5

110 100 101

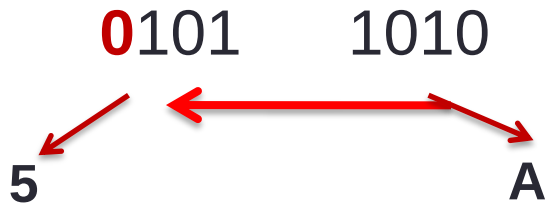
$(6\ 4\ 5)_8 = (110\ 100\ 101)_2$

OCTAL	BINAIRE
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

3.6 From base 2 to base 16

Hexadecimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

- divide the number into groups of 4 bits and rep
  group with its hexadecimal value (starting from
- If the number of bits is not a multiple of 4, com
  0s on the left.
- **Example** : Convert $(1011010)_2$ to base 16



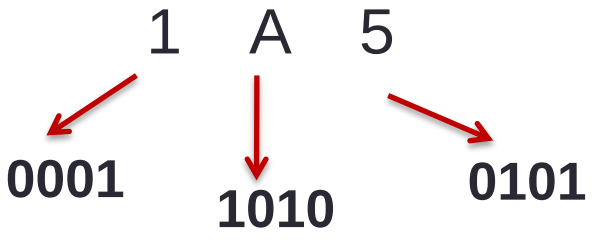
$(0101\ 1010)_2 = (5\ A)_{16}$

Hexadecimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

3.7 From base 16 to base 2

- simply transcribe each digit of this number into digits starting from the left.

- **Example** : Convert $(1A5)_{16}$ to base 2



$(1\ A\ 5)_{16} = (0001\ 1010\ 0101)_2$

Hexadecimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

- To convert fractional numbers, we separate the int from the fractional part.
- Integer part is treated as described above.
- The decimal part is converted from left to right.

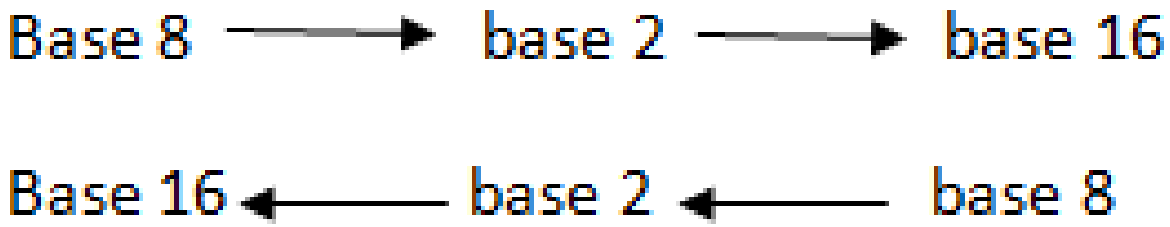
Example:

- $(11101,01011)_2 = (011\ 101\ ,\ 010\ 110)_2 = (\ 35,26)_8$
- $(11101,01011)_2 = (0001\ \ 1101\ ,\ 0101\ 1000)$
  $= (1D,58)_{16}$

Hexadecimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
F	1111

OCTAL	BINARY
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

4.8 From base 8 to base 16 or vice versa



Example : Convert $(232)_8$ to base 16

$(2\ 3\ 2)_8 = (010\ \ 011\ \ 010)_2 = (0000/1001/1010)_2 = (9\ A)_{16}$

- **APPLICATION**

- $(35)_{10} = (N)_2$

N=100011 (succ DIV)

- $(1010001)_2 = (N)_{16}$

N=51 (4bits)

- $(17)_8 = (N)_2$

N=001 111 (Transcribe 3bits)

- $(2A)_{16} = (N)_2$

N=0010 1010 (Transcribe 4bits)

- $(17)_8 = (N)_{10}$

N= 15 (Polynomial)